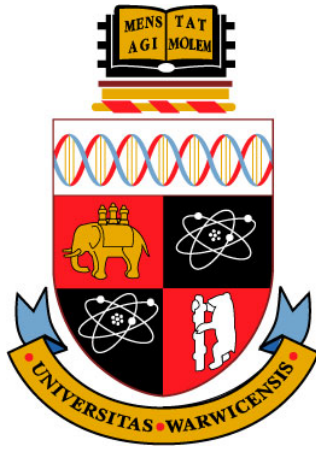




Residual Learning for Robust MPC Under Model Discrepancy: An Empirical Study on an Inverted Pendulum

Summer Research Project

Markos Hajivassiliou

Supervisor: Dr Mona Faraji Niri

Department of Engineering

University of Warwick

2026

Abstract

Model Predictive Control (MPC) relies on an accurate model of the system it controls, yet many real-world systems are difficult or impossible to model exactly. Residual learning — using a neural network to correct a known-imperfect nominal model — is one proposed remedy, but the practical range of model error over which this correction is actually worthwhile is rarely characterized empirically. This report investigates that question directly, on a simple inverted pendulum, by deliberately introducing five severities of nominal-model length error (5–75%) and comparing, at each severity, a robust (multi-stage) MPC controller using the discrepant model alone against the same controller augmented with a residual neural network trained to correct it. Rather than deriving a new theoretical stability guarantee, the study applies the stability framework of Learning-Based Model Predictive Control (LBMPC), implementing a simplified necessary-condition feasibility check in place of its full invariant-set construction, and relying on direct closed-loop simulation as the primary evidence of practical stability. Residual learning is found to provide no benefit, and a small cost, at low discrepancy; substantial, unambiguous benefit at moderate-to-high discrepancy, recovering cases in which the uncorrected controller fails outright; and partial, incomplete benefit at the most severe discrepancy tested — a non-monotonic pattern corroborated independently by the network’s own prediction accuracy. A supplementary finding shows that the specific robustness margin used contributed negligibly to the observed improvements relative to the residual correction itself. These results offer an honestly-scoped empirical characterization of when learning-enhanced control is, and is not, worthwhile.

Contents

Abstract	ii
1 Introduction	1
2 Literature Review	2
2.1 Learning-Based MPC (LBMPC)	2
2.2 Closest Prior Implementation	2
2.3 Black-Box Alternatives	2
2.4 Contribution	2
3 Methodology	3
3.1 System Model	3
3.2 Nominal MPC Baseline	3
3.3 Inducing Model Discrepancy	4
3.4 Data Generation	4
3.5 Residual Neural Network	4
3.6 Terminal Ingredients	5
3.7 Uncertainty Bound and Simplified Feasibility Check	6
3.8 Robust Model Predictive Controller	6
3.9 Plain MPC Baseline and Margin Sensitivity Check	6
3.10 Closed-Loop Evaluation	6
4 Results	7
4.1 Residual Learnability	7
4.2 Closed-Loop Stability	8
4.3 Simplified Feasibility Check	11
5 Discussion	12
5.1 The Benefit of Residual Learning Is Not Uniform	12
5.2 The Robustness Margin Contributed Negligibly at the Tested Setting	12
5.3 The Simplified Feasibility Check Understates Real-World Failure	13
5.4 Limitations	13
5.5 Future Work	13
6 Conclusion	14

1 Introduction

Model Predictive Control (MPC) is a control strategy that, at every time step, uses a model of the system to predict its behaviour over a short future horizon, solves an optimization problem to choose the best sequence of control actions over that horizon, and applies only the first action before repeating the process at the next time step with updated information. This repeated re-planning gives MPC a degree of natural robustness to model imperfections, since it continuously corrects its plan based on what actually happens, rather than committing to a single open-loop plan in advance.

Model Predictive Control (MPC) relies on an accurate model of the system it is controlling. In many real-world settings, obtaining such a model exactly is difficult or impossible, the true dynamics may be too complex, or key physical parameters (mass, length, friction) may be only approximately known. One approach to this problem is *learning-based MPC*: using a machine learning model to learn a correction to an imperfect nominal model, rather than relying on the nominal model alone.

This project investigates a specific, practical question: **when a robust MPC controller is built around a deliberately incorrect nominal model, does adding a neural-network-learned correction (residual learning) meaningfully improve closed-loop stability, and if so, over what range of model error?**

We study this question on a simple inverted pendulum, a canonical benchmark system in control theory. We deliberately introduce error into the controller’s belief about the pendulum’s length, at five severities (5%, 15%, 30%, 50%, and 75% underestimation), and compare three conditions at each severity: a nominal MPC with the wrong system dynamics, a robust MPC using the incorrect nominal model alone, and the same robust MPC augmented with a neural network trained to predict and correct the resulting model error.

Rather than deriving a new theoretical stability guarantee, we adopt and apply the stability framework of Learning-Based Model Predictive Control (LBMPC) [2] in which safety is derived from a nominal model with bounded uncertainty, decoupled from the accuracy of any learned correction. We state clearly where we simplify this framework relative to its full theoretical treatment, and rely on direct closed-loop simulation as our primary evidence of practical stability.

Our central finding is that residual learning’s benefit is **not uniform** across the range of model error we tested: it offers little or no benefit at low discrepancy (5–15%), where there is little error to correct; it provides substantial, clear benefit at moderate-to-high discrepancy (30–50%), successfully stabilizing a case where the uncorrected controller fails outright; and at the most extreme level we tested (75%), it improves but does not fully resolve the failure, with the corrected system settling into a damped oscillation rather than either full recovery or complete failure.

2 Literature Review

Learning-based MPC spans a range of approaches, well summarized by Hewing et al.[3], who categorizes the field by what is learned (a full dynamics model, a residual correction, or a control policy) and how safety guarantees are preserved. This work is situated in the *residual learning* branch of that taxonomy: rather than learning the full system dynamics from scratch, we assume access to a nominal, but incorrect, model, and learn only the correction needed to account for its error.

2.1 Learning-Based MPC (LBMPC)

The theoretical framing adopted here follows Aswani, Gonzalez, Sastry, and Tomlin’s LBMPC framework[2], which decouples safety from performance: a nominal model with a bounded uncertainty set provides a deterministic stability guarantee, independent of how, or how well, a second, learned “oracle” model corrects it. This decoupling is what permits a *static*, offline-trained neural network (rather than LBMPC’s original online-adaptive oracle) to remain a valid instantiation of the same framework.

2.2 Closest Prior Implementation

Anderson’s report on LBMPC for an inverted pendulum[1] is the closest prior implementation to this study: it applies the redreference here and above LBMPC framework to the same benchmark system and empirically compares LQR, robust MPC, and LBMPC. This work extends that comparison in two respects: it systematically varies the *severity* of nominal-model error across five discrepancy levels rather than a single case, and it pairs each level with a direct closed-loop comparison against the same robust controller *without* the learned correction, isolating the correction’s specific contribution to stability.

2.3 Black-Box Alternatives

A separate line of work learns the full system dynamics as a black box rather than a residual correction on a nominal model, with accompanying stability guarantees (e.g.[5] and [4]) Residual learning is adopted here instead, as it requires substantially less training data to achieve comparable accuracy and aligns directly with the LBMPC theoretical framework this study builds on.

2.4 Contribution

This work provides a systematic, quantified characterization of the range of nominal-model error over which residual learning improves closed-loop stability under a fixed robust MPC framework, via a direct paired comparison across five discrepancy severities. A comparison not present in the prior work above.

3 Methodology

3.1 System Model

The system under study is a simple, directly-actuated inverted pendulum: a point mass at the end of a rigid arm, with a torque applied at the pivot.

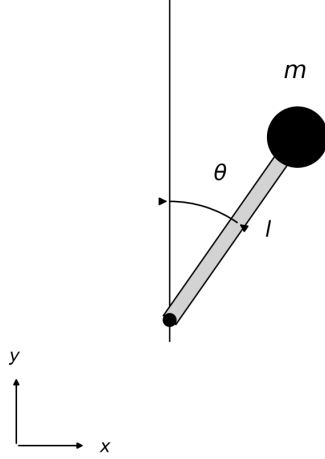


Figure 1: Schematic of the inverted pendulum system and angle convention used throughout this report.

The continuous-time dynamics are given by

$$\ddot{\theta} = \frac{mgl \sin(\theta) + u - b\dot{\theta}}{ml^2} \quad (3.1)$$

where θ is the pendulum's angle from upright ($\theta = 0$, the unstable equilibrium), $\dot{\theta}$ is the angular velocity, u is the applied torque, m is the mass, l is the pendulum length, b is a damping coefficient, and g is gravitational acceleration. Defining the state $x = [\theta, \dot{\theta}]^\top$, Equation 3.1 is written in state-space form as $\dot{x} = f(x, u)$.

The continuous-time dynamics are discretized using 4th-order Runge-Kutta integration with a fixed time step $\Delta t = 0.02$ s, yielding a discrete-time update $x_{k+1} = F(x_k, u_k)$, used throughout the remainder of this study. The parameters used are $m = 1.0$ kg, $l = 0.5$ m, $b = 0.1$, and $g = 9.81$ m/s².

3.2 Nominal MPC Baseline

A standard MPC controller is first implemented using the exact (true) system dynamics, to validate the modelling, cost function, and constraint setup before introducing any model error or learning. At each time step k , this controller solves the following finite-horizon optimal control problem:

$$\begin{aligned}
& \min_{u_0, \dots, u_{N-1}} && x_N^\top P x_N + \sum_{i=0}^{N-1} (x_i^\top P x_i + r u_i^2) && (3.2) \\
\text{subject to} &&& x_{i+1} = F(x_i, u_i), \quad i = 0, \dots, N-1 \\
&&& x_0 = x(k) \\
&&& u_{\min} \leq u_i \leq u_{\max}, \quad i = 0, \dots, N-1
\end{aligned}$$

where N is the prediction horizon, P is a terminal cost matrix (Section 3.6), r is a scalar weight penalizing control effort, and $x(k)$ is the current state. Only the first control action, u_0 , is applied to the system; the optimization is then re-solved at the next time step using the updated state, the standard *receding horizon* principle that gives MPC its characteristic feedback behaviour and inherent tolerance to model imperfections. In this study, $N = 20$, $r = 0.1$, and $u_{\min} = -u_{\max} = -5 \text{ Nm}$. This nominal controller successfully stabilizes the pendulum from a range of initial conditions and serves as the source of training data described below.

The horizon length N and control-effort weight r were selected through empirical tuning: values were chosen such that the nominal controller (true dynamics, no discrepancy) converged smoothly to the upright equilibrium without excessive control effort or oscillatory behaviour, and were then held fixed across all subsequent experiments to ensure a consistent basis for comparison across discrepancy levels and conditions.

3.3 Inducing Model Discrepancy

To study the effect of model error, a *nominal* model is constructed that consistently underestimates the pendulum’s true length by a fixed fraction δ :

$$l_{\text{wrong}} = l_{\text{true}} (1 - \delta) \quad (3.3)$$

Five discrepancy severities are tested: $\delta \in \{0.05, 0.15, 0.30, 0.50, 0.75\}$, while the true simulated system retains its correct length throughout. This choice isolates a single, interpretable source of model error whose severity can be precisely controlled, rather than introducing multiple simultaneous sources of uncertainty.

3.4 Data Generation

Training data is generated by simulating the true dynamics under randomized torque excitation, sampled uniformly from $u \sim \mathcal{U}(-u_{\max}, u_{\max})$ at each time step, from a spread of initial conditions covering both near-equilibrium and moderately displaced states. This yields a dataset of (x_k, u_k, x_{k+1}) triples, used both to train the residual network (Section 3.5) and to compute the empirical uncertainty bound (Section 3.7).

3.5 Residual Neural Network

For each discrepancy level, the true residual is computed for every data point as

$$r_k = x_{k+1} - F_{\text{wrong}}(x_k, u_k) \quad (3.4)$$

where F_{wrong} denotes the discrete dynamics computed using l_{wrong} in place of l_{true} . A feedforward neural network is trained, independently per discrepancy level, to predict this residual from the current state and input. The network has the form

$$\hat{r} = W_3 \sigma(W_2 \sigma(W_1[\theta, \dot{\theta}, u]^\top + b_1) + b_2) + b_3 \quad (3.5)$$

with two hidden layers of 32 units each, $\sigma(\cdot) = \max(0, \cdot)$ (the ReLU activation function), and weight matrices $W_1 \in R^{32 \times 3}$, $W_2 \in R^{32 \times 32}$, $W_3 \in R^{2 \times 32}$. Each network is trained on an 80/20 train/validation split of the dataset described in Section 3.4, using the Adam optimizer to minimize mean squared error between $\hat{r}$ and r_k . Validation RMSE is reported as the network's empirical accuracy at each discrepancy level.

3.6 Terminal Ingredients

To connect the empirical setup to the LBMPC theoretical framework [2], the terminal ingredients required for its stability argument are constructed as follows. The nominal (discrepant) dynamics are linearized about the upright equilibrium $(\theta, \dot{\theta}, u) = (0, 0, 0)$ via the Jacobians

$$A = \left. \frac{\partial f_{\text{wrong}}}{\partial x} \right|_{(0,0,0)}, \quad B = \left. \frac{\partial f_{\text{wrong}}}{\partial u} \right|_{(0,0,0)} \quad (3.6)$$

computed via automatic differentiation. As a concrete example, at $\delta = 0.30$ ($l_{\text{wrong}} = 0.35$ m), this yields

$$A = \begin{bmatrix} 0 & 1 \\ 28.03 & -0.816 \end{bmatrix}, \quad B = \begin{bmatrix} 0 \\ 8.163 \end{bmatrix} \quad (3.7)$$

A terminal cost matrix P and terminal feedback gain K are then obtained by solving the discrete algebraic Riccati equation (DARE),

$$P = A^\top P A - A^\top P B (R + B^\top P B)^{-1} B^\top P A + Q \quad (3.8)$$

with $Q = I_2$ and $R = 1$, and the corresponding gain

$$K = -(R + B^\top P B)^{-1} B^\top P A \quad (3.9)$$

For every discrepancy level tested, the resulting closed-loop matrix $A + BK$ was confirmed to have all eigenvalues strictly inside the unit circle, verifying K is stabilizing for the linearized system. A terminal set is defined as a conservatively small sublevel set of the resulting quadratic cost,

$$\Omega = \{x : x^\top P x \leq \alpha\}, \quad \alpha = 0.05 \quad (3.10)$$

chosen without a rigorous derivation of its maximal valid size (see Chapter 5, Limitations).

3.7 Uncertainty Bound and Simplified Feasibility Check

The LBMPC stability guarantee requires a bound, W , on the nominal model’s uncertainty. This is estimated empirically as the maximum observed residual, per state dimension, over a dataset restricted to a bounded neighbourhood of the equilibrium ($|\theta| < 0.4$, $|\dot{\theta}| < 1.0$):

$$W = \max_k |r_k|, \quad \text{for } k \text{ such that } |\theta_k| < 0.4, |\dot{\theta}_k| < 1.0 \quad (3.11)$$

As a practical, necessary (but not sufficient) proxy for the full invariant-set feasibility analysis prescribed by the theory, we check whether the terminal controller’s corrective action for this worst-case disturbance remains within the actuator’s torque limit:

$$\|K W\| < u_{\max} \quad (3.12)$$

3.8 Robust Model Predictive Controller

The controller is implemented as a multi-stage (scenario-based) robust MPC, in which the nominal model’s length parameter is treated as uncertain and represented by three discrete scenarios at the first branching step of the horizon:

$$l \in \{l_{\text{wrong}}, l_{\text{wrong}}(1+w), l_{\text{wrong}}(1-w)\}, \quad w = 0.05 \quad (3.13)$$

The controller’s terminal cost uses the Riccati-derived matrix P for the corresponding discrepancy level. In the NN-corrected condition, the residual network’s prediction (Equation 3.5), reconstructed symbolically to allow integration with the optimization framework, is added to the nominal model’s prediction at every step of the controller’s internal dynamics.

3.9 Plain MPC Baseline and Margin Sensitivity Check

To isolate the specific contribution of the multi-stage robustness mechanism from that of the residual correction, a plain (single-scenario, i.e. Equation 3.13 evaluated only at l_{wrong}) MPC using the same discrepant nominal model is also evaluated at each discrepancy level. The multi-stage implementation’s correctness is additionally verified by testing a substantially wider margin ($w = 0.30$) at one discrepancy level, confirming the mechanism produces measurably different behaviour when the margin is large enough to matter.

3.10 Closed-Loop Evaluation

For each of the five discrepancy levels, three closed-loop simulations of 18 seconds (900 steps) are run: the plain MPC baseline, the robust MPC using the discrepant nominal model alone, and the robust MPC with the residual correction added. All three are evaluated against the same true simulated dynamics, from the same initial condition ($\theta = 0.3 \text{ rad}$, $\dot{\theta} = 0 \text{ rad/s}$). This yields fifteen trajectories in total, forming the basis of the results presented in Chapter 4.

4 Results

4.1 Residual Learnability

Figure 2 shows the trained neural network’s validation RMSE at each discrepancy level. Prediction error increases gradually through 50% discrepancy (RMSE 0.012 to 0.077), then rises sharply at 75% (RMSE 0.339), a roughly 4–5x jump compared to the increase seen at any prior step. This suggests 75% discrepancy represents a qualitatively harder regime for residual learning, not simply a continuation of the trend observed at lower severities.

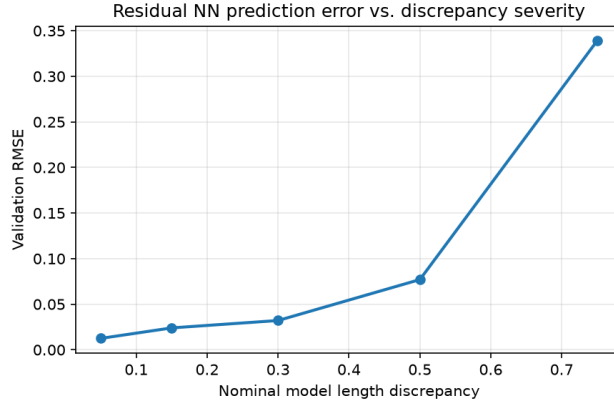


Figure 2: Residual neural network validation RMSE as a function of nominal model length discrepancy.

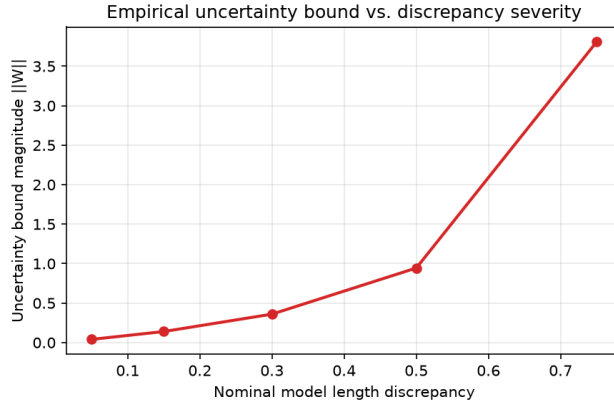


Figure 3: Empirical uncertainty bound magnitude ($\|W\|$) as a function of nominal model length discrepancy.

Figure 3 shows the corresponding trend in the empirically estimated uncertainty bound W (Equation 3.11), computed independently of the neural network. The same qualitative pattern is observed, gradual growth through 50%

discrepancy, followed by a sharp increase at 75%, providing a second, independent confirmation that 75% discrepancy represents a qualitatively distinct regime, consistent with the residual network’s validation error.

4.2 Closed-Loop Stability

Figure 4 shows the closed-loop trajectory of the pendulum angle (θ) over an 18-second window, for three conditions at each of five discrepancy levels: a plain MPC using the discrepant nominal model, a robust (multi-stage) MPC using the same model, and the robust MPC augmented with the residual neural network correction.

Across all five levels, the plain and robust (nominal-only) conditions are visually indistinguishable, indicating the multi-stage uncertainty scenarios ($w = 0.05$) contributed negligibly to closed-loop behaviour at this margin size; this is addressed further in Chapter 5.

The residual-corrected condition shows a distinct, non-monotonic pattern across discrepancy severity:

- **At 5% and 15% discrepancy**, the NN-corrected trajectory converges more slowly than the uncorrected baseline, settling at a small but non-zero final angle (≈ 0.02 – 0.14 rad) where the uncorrected baseline converges essentially to zero.
- **At 30% discrepancy**, the uncorrected baseline shows little further progress toward the target over the full simulation window, remaining near 0.13 – 0.15 rad; the NN-corrected trajectory continues to converge throughout and crosses the target angle entirely, settling at ≈ -0.14 rad, a qualitative difference in convergence behaviour, though both final deviations remain modest in absolute terms.
- **At 50% discrepancy**, the uncorrected baseline diverges completely, settling near $\theta \approx 2.28$ rad; the NN-corrected controller converges cleanly to near zero. The clearest single demonstration of the correction’s benefit.
- **At 75% discrepancy**, the uncorrected baseline settles at the downward equilibrium ($\theta \approx \pi$); the NN-corrected controller does not reach upright, but converges via a damped oscillation to an intermediate angle (≈ 1.5 – 1.75 rad). A partial, but incomplete, improvement.

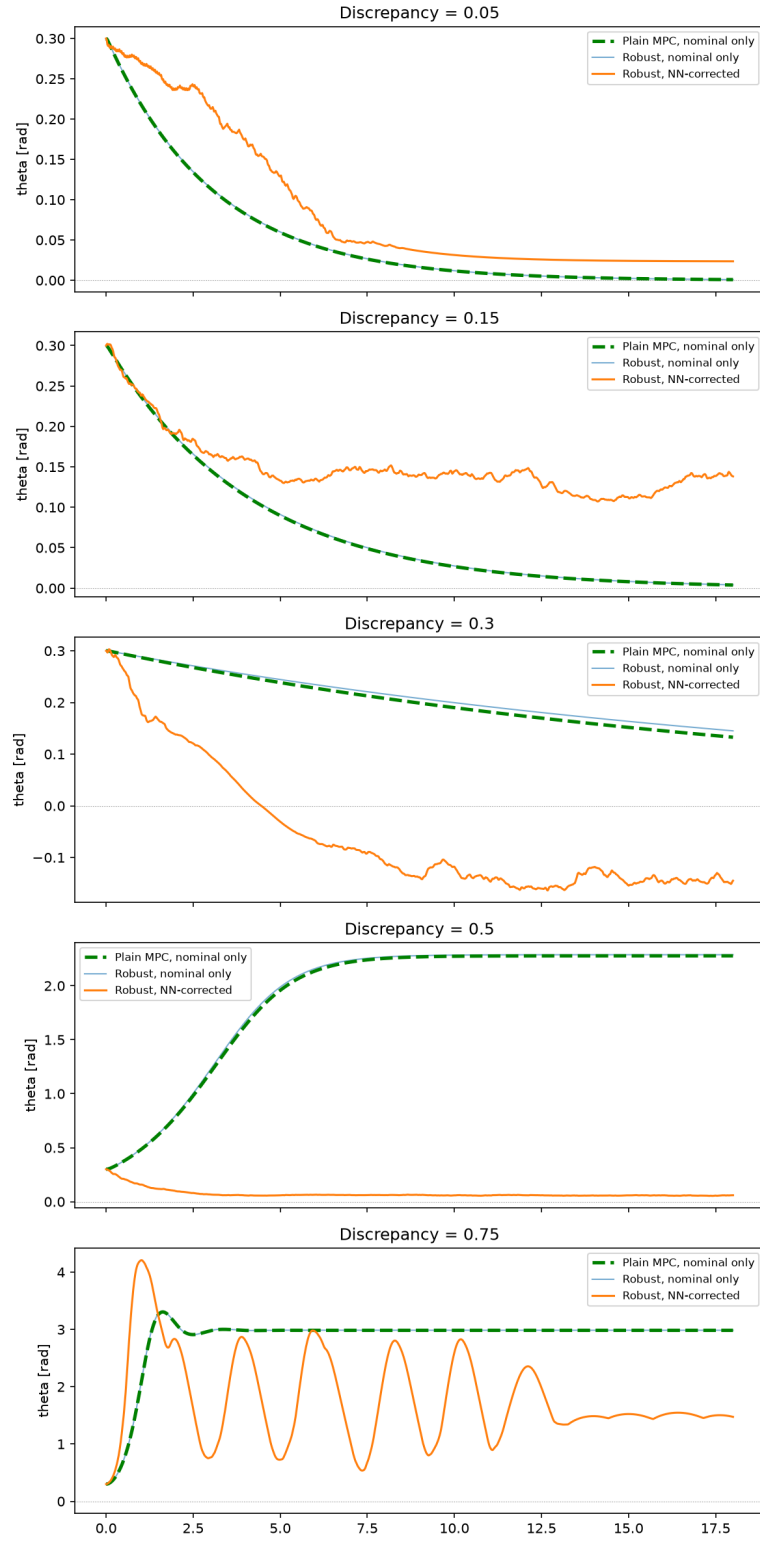


Figure 4: Closed-loop pendulum angle over 18 seconds, at five discrepancy levels, for the plain MPC, robust nominal-only, and robust NN-corrected conditions.

Table 1: Summary of empirical results across discrepancy levels. θ_{final} values are the pendulum angle (rad) at $t = 18\text{s}$; smaller magnitude indicates closer approach to the upright target ($\theta = 0$). All five discrepancy levels were found feasible under the necessary-condition check (Section 4.3).

Discrepancy	RMSE	θ_{final} (Plain)	θ_{final} (Ro-bust)	θ_{final} (NN-corrected)
5%	0.0125	0.0009	0.0009	0.0236
15%	0.0240	0.0038	0.0040	0.1382
30%	0.0321	0.1328	0.1451	-0.1449
50%	0.0770	2.2768	2.2871	0.0595
75%	0.3389	2.9845	2.9862	1.4736

Figure 5 summarizes the final angular deviation at $t = 18\text{s}$ across all three conditions and discrepancy levels.

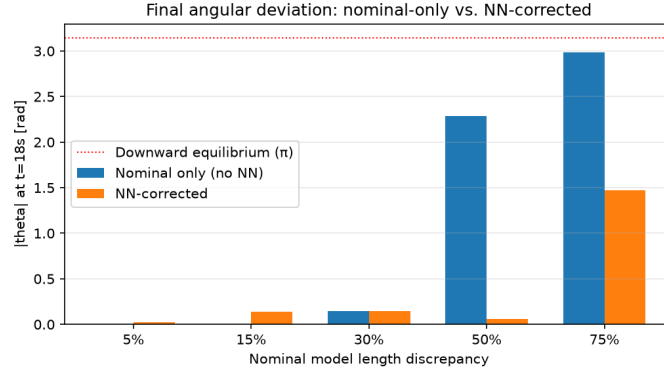


Figure 5: Final angular deviation at $t = 18\text{s}$, nominal-only vs. NN-corrected, across all five discrepancy levels.

Figure 6 shows the phase-space trajectory (θ vs. $\dot{\theta}$) for the 75% discrepancy case specifically, illustrating the qualitative difference between the uncorrected controller’s single-arc convergence to the downward equilibrium and the NN-corrected controller’s damped oscillatory approach to an intermediate point.

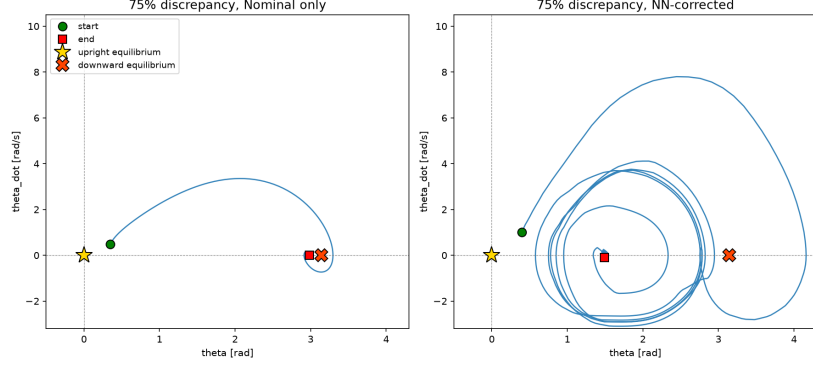


Figure 6: Phase portrait (θ vs. $\dot{\theta}$) at 75% discrepancy: uncorrected (left) vs. NN-corrected (right).

4.3 Simplified Feasibility Check

Applying the necessary-condition feasibility check described in Section 3.7 (Equation 3.12), all five discrepancy levels were found feasible, including 75%, where the closed-loop simulation nonetheless failed to reach the upright target. This discrepancy between the simplified check’s prediction and the observed closed-loop outcome is addressed in Chapter 5.

5 Discussion

5.1 The Benefit of Residual Learning Is Not Uniform

The central finding of this study is that residual learning’s practical value depends strongly on the severity of the underlying model discrepancy. At low discrepancy (5–15%), there is little systematic error for the network to correct, and the learned correction appears to introduce a small amount of prediction noise that mildly *slows* convergence relative to the uncorrected baseline. A modest but consistent negative result. At moderate-to-high discrepancy (30–50%), the correction provides substantial, unambiguous benefit, most clearly at 50%, where it is the difference between complete failure (divergence to the wrong equilibrium) and clean convergence to the target. At the most extreme level tested (75%), the correction meaningfully improves the outcome, replacing a static failure at the downward equilibrium with a damped oscillation toward an intermediate angle, but does not fully resolve it.

This pattern is consistent with, and corroborated by, the residual network’s own validation error (Figure 2): the sharp increase in RMSE between 50% and 75% discrepancy directly foreshadows the point at which closed-loop performance also transitions from full recovery to partial recovery. The consistency between these two independently measured quantities, open-loop prediction accuracy and closed-loop control outcome strengthens confidence that both are capturing a genuine property of the system, rather than measurement noise.

5.2 The Robustness Margin Contributed Negligibly at the Tested Setting

The near-identical performance of the plain and robust (multi-stage) MPC formulations, observed across all five discrepancy levels (Figure 4), might suggest a flaw in the robust implementation. This was verified not to be the case: a supplementary test using a substantially wider uncertainty margin ($w = 0.30$, rather than $w = 0.05$ used throughout the main study) produced a measurably different, and more conservative, closed-loop outcome, confirming the multi-stage mechanism responds correctly to its configured margin.

The negligible effect observed at $w = 0.05$ instead reflects the relationship between that margin and the discrepancy severities under study: because $w = 0.05$ is small relative to discrepancies of up to 75%, the three scenarios considered by the optimizer at each solve (Equation 3.13) are close enough to one another that little genuine disagreement exists for the controller to hedge against. As a result, ordinary MPC feedback, which both the plain and robust formulations share. Accounts for the large majority of the stabilization observed in the nominal-only baseline, and the residual correction, not the robustness margin, is the primary source of the improvements reported above. This finding is itself useful: it isolates residual learning as the dominant contributing factor in this study’s results, independent of the specific robust MPC formulation’s design.

5.3 The Simplified Feasibility Check Understates Real-World Failure

The necessary-condition feasibility check (Equation 3.12) found all five discrepancy levels technically feasible under actuator constraints, including 75% , a level at which the closed-loop simulation clearly fails to reach the control objective. This gap illustrates a known limitation of the simplified check: it evaluates only a single worst-case correction at one instant, and does not account for disturbances compounding over the full prediction horizon, nor for the system’s actual nonlinear behaviour far from the equilibrium about which the check’s linear quantities (K , P) were derived. The closed-loop simulation, not the simplified check, should be treated as the primary evidence of practical stability in this study.

5.4 Limitations

This study makes several deliberate simplifications relative to the full LBMPC theoretical framework, made explicit here for transparency:

- The terminal set size (α) was chosen conservatively rather than rigorously derived from the nonlinear system’s Taylor-remainder error bound.
- The uncertainty bound (W) was estimated empirically from a dataset restricted to a bounded neighbourhood of the equilibrium, rather than derived analytically.
- The full disturbance-invariant set construction required for LBMPC’s formal stability guarantee was not implemented; the necessary-condition feasibility check used here is a practical proxy, not a substitute for the complete guarantee.
- The residual network’s uncertainty bound (used to justify the robust formulation) was held fixed across the NN-corrected and nominal-only conditions, rather than reduced to reflect the network’s improved accuracy; this may understate the potential benefit of combining residual learning with an appropriately-tuned robustness margin.
- Results are reported for a single initial condition ($\theta = 0.3$ rad); generalization to other starting states was not tested.

5.5 Future Work

Natural extensions include: deriving the terminal set size and uncertainty bound analytically rather than conservatively; implementing the full tube-MPC invariant-set construction to obtain a rigorous rather than necessary-condition-only stability guarantee; extending the residual learning approach to a second, more complex nonlinear system; and exploring an online-adaptive variant of the residual correction, in which the network continues to update using data collected during closed-loop operation, as in the original LBMPC formulation[2].

6 Conclusion

This project set out from a practical observation: real systems are often too complex, or too poorly characterized, to model exactly, and learning-based correction is frequently proposed as a remedy. What is less often established empirically is *where the boundaries of that remedy actually lie*. At what point does a learned correction genuinely rescue a poorly-modelled controller, and at what point does the underlying model error simply exceed what learning can practically compensate for. This study set out to answer that question directly, for a concrete, tractable system, rather than assuming the answer is either always or never.

A complete pipeline was built and validated to make this question answerable: a nonlinear pendulum simulator, a nominal MPC baseline, a controlled discrepancy-injection procedure, a residual-learning dataset-generation and training pipeline, a theoretically-grounded set of terminal ingredients connecting the empirical setup to the LB MPC stability framework, and a robust multi-stage MPC controller capable of testing both corrected and uncorrected conditions under closed-loop simulation. Each simplification made relative to the full theoretical treatment. The terminal set size, the uncertainty bound estimation, the necessary-condition feasibility check in place of the full invariant-set construction, was made explicitly and justified on its own terms, rather than left implicit.

The resulting empirical answer is that residual learning’s value is genuinely conditional, not universal. It offered no benefit, and a small cost, when the underlying model error was already small, it provided decisive, unambiguous benefit once that error grew large enough to threaten stability outright and its benefit diminished, though did not vanish, once the error became severe enough to strain the correction task itself. A transition foreshadowed independently by the network’s own prediction accuracy, lending confidence that this is a genuine property of the system rather than an artefact of any single experiment. A supplementary, directly verified finding further showed that the specific robustness margin adopted in this study contributed little on its own, isolating residual learning, rather than the robust formulation’s structure, as the dominant factor behind the improvements observed.

Taken together, this suggests that the practical question worth asking of a learning-enhanced controller is not simply *does learning help*, but *over what range of model error does it help*, and by how much. A question this study answers concretely, if narrowly, for one system, and which the future work outlined in Section 5.5 could extend toward greater rigour and generality.

References

- [1] B. Anderson. Inverted pendulum stabilization via learning-based model predictive control, 2019. Project report.
- [2] A. Aswani, H. Gonzalez, S. S. Sastry, and C. Tomlin. Provably safe and robust learning-based model predictive control. *Automatica*, 49(5):1216–1226, 2013.
- [3] L. Hewing, K. P. Wabersich, M. Menner, and M. N. Zeilinger. Learning-based model predictive control: Toward safe learning in control. *Annual Review of Control, Robotics, and Autonomous Systems*, 3:269–296, 2020.
- [4] P. K. Mishra, M. V. Gasparino, A. E. B. Velasquez, and G. Chowdhary. Deep model predictive control with stability guarantees, 2021. arXiv:2104.07171.
- [5] I. Schimperna and L. Magni. Robust constrained nonlinear model predictive control with gated recurrent unit model. *Automatica*, 161:111472, 2024.